DreamBooth

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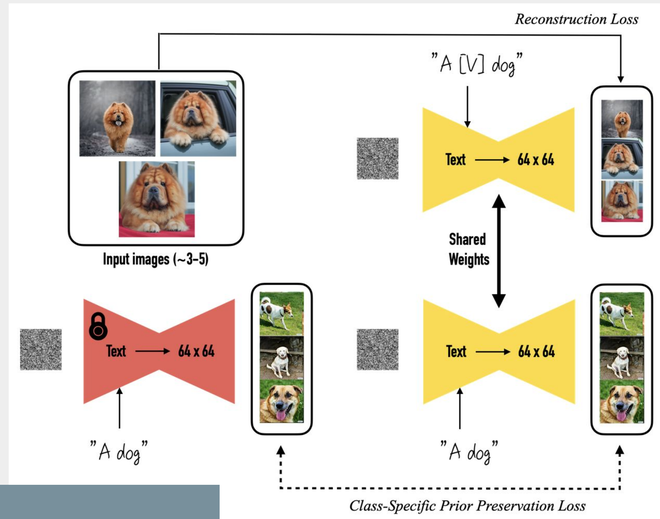
Motivation

The goal: Generate novel images with only a few examples

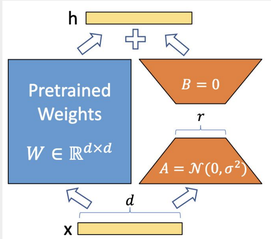
The main challenge here is preserving **unique features of a subject** without **overfitting** the input images.

Methods



To accomplish our goal, we fine-tuned **Stable Diffusion 1.5**. Our objective is to generate output similar to our examples when a **special adjective token** is present, but still generate **diverse examples of the subject's class** otherwise.

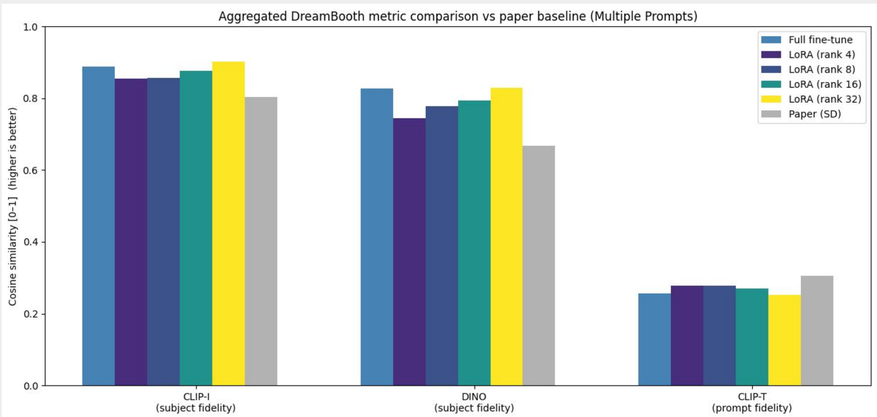
We measure the **quality** of the trained model's output by comparing embeddings of both the input images **(DINO & CLIP-I)** and prompts **(CLIP-T)** with the output.



We also added **LoRA** (Low Rank Adaptation) to reduce memory usage. LoRA allows us to fine-tune by adjusting **pairs of smaller matrices** instead of the huge weight matrices from the pre-trained model.

Results

We were able to achieve results **comparable to and even better** than the original paper in some cases. We also tried testing with a **human** subject, and although the fidelity was worse, we still got decent results.



Rank = 4

Rank = 32

LoRA has only a small negative impact on fidelity.
However, too low of a rank appears to hurt overall quality.

Conclusion

- **Careful fine-tuning** can very effectively add to the capabilities of a model while retaining abilities that would be expensive or difficult to train from scratch.
- There are numerous **tradeoffs** in what to optimize a model for, such as **quality, fidelity, scene diversity**, and more.
- Future work to explore how well this method can generalize to **learning multiple subjects, more complicated types of subjects**, etc. would be useful to help apply this technique in more ways to real-world applications.

References

- Ruiz, Nataniel, et al. 'DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation'. arXiv [Cs.CV], 2023, arxiv.org/abs/2208.12242. arXiv.
- Hu, Edward J., et al. 'LoRA: Low-Rank Adaptation of Large Language Models'. arXiv [Cs.CL], 2021, arxiv.org/abs/2106.09685. arXiv.